ADAS Sensor Fusion - High-Level Design

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1. Component Overview

This document describes the Sensor Fusion software responsible for combining radar and camera data for advanced drive

Software Components

Component: SensorFusionComponent
Component: RadarProcessingComponent
Component: CameraProcessingComponent

2. Interfaces and Ports

Interface: RadarDataInterface
Interface: CameraDataInterface
Interface: FusedObjectListInterface
PPort: SensorFusion_PPort
RPort: RadarProcessing_RPort
RPort: CameraProcessing_RPort
SensorFusionComponent requires RadarDataInterface via RadarProcessing_RPort and CameraDataInterface via Camera

3. Signals

Signal: RadarObjectList_Signal
Signal: CameraObjectList_Signal
Signal: FusedObjectList_Signal

4. Dependencies

SensorFusionComponent depends on RadarProcessingComponent and CameraProcessingComponent for raw object lists

5. Functional Flow

RadarProcessingComponent and CameraProcessingComponent independently publish object lists. SensorFusionCompon